

Date: October 30, 2025

Scope: home.py, shapes.py, board.py, game.py, login.py, leaderboard.py

Component testing report

Results

- Components Tested: 5
- Test Scenarios Executed: 24
- Tests Passed: 18
- Tests Failed: 6
- Critical Issues: 4

Components status

- **Stable Components:** home.py, login.py
- **Components with Minor Issues:** shapes.py, board.py
- **Components Needing Critical Fixes:** game.py

Issues:

1. **Import chain broken:** Game cannot start due to missing module imports
2. **No game-over condition:** Game continues when blocks reach top border
3. **No rotation cooldown:** Shape rotates uncontrollably fast
4. **State persistence bug:** Failed game state carries over between sessions
5. **Leaderboard missing:** Score cannot be saved
6. **Input sensitivity:** Movement lacks proper throttling

Component Readiness

home.py: Ready for Integration

- **Public Interface: Stable**
- **Integration: Good**
- **Issues: None critical**
- **Confidence: High**

login.py: Ready for Integration

- **Public Interface: Stable**
- **Integration: Good**
- **Issues: None critical**
- **Confidence: High**

shapes.py: Ready with Minor Issues

- **Public Interface: Stable**
- **Integration: Good**
- **Issues: Minor scoring conflicts with game.py**
- **Confidence: High - Core logic solid**

board.py: Ready with Minor Issues

- **Public Interface: Stable**
- **Integration: Good**
- **Issues: Grid rendering offset issues**
- **Confidence: High - Rendering reliable**

game.py: Requires Critical Fixes

- **Public Interface: Broken**
 - **Integration: Poor**
 - **Issues: 4 critical gameplay flaws**
 - **Confidence: Low - Core gameplay broken**
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Quality Assessment

Component	Code Quality	Integration	Documentation	Overall
home.py	Good	Good	Partial	Good
login.py	Good	Good	Partial	Good
shapes.py	Good	Good	Partial	Good
board.py	Good	Good	Partial	Good
game.py	Poor	Poor	Partial	Poor

Component Defect Log

CRITICAL DEFECTS (PRIORITY 1)

ID	Component	Description	Severity
D-31	game.py	Import chain broken: Game cannot start	HIGH
D-32	game.py	No game-over detection when board fills	HIGH
D-33	game.py	Game state persists causing visual corruption	HIGH
D-34	game.py	No rotation cooldown - uncontrollable spinning	HIGH

MAJOR DEFECTS (PRIORITY 2)

ID	Component	Description	Severity
D-35	game.py	Leaderboard module missing - scores not saved	MEDIUM
D-36	game.py	Input sensitivity too high - movement lacks throttle	MEDIUM
D-37	All	Resource loading fails silently	MEDIUM

MINOR DEFECTS (PRIORITY 3)

ID	Component	Description	Severity
D-38	shapes.py	Scoring conflicts with game.py logic	LOW

Test Results Summary

home.py Tests: 4/4 PASSED

- All button functionality tests passed
- UI components stable and reliable

login.py Tests: 4/4 PASSED

- Input handling tests passed
- User interface tests passed

shapes.py Tests: 5/6 PASSED

- **Shape movement and rotation**
- **Collision detection**
- **Line clearing logic**
- **Scoring within module**
- **FAILED:** Scoring integration test (conflicts with game.py)

board.py Tests: 4/5 PASSED

- **Grid initialization**
- **Dimension calculations**
- **UI rendering basics**
- **SKIPPED:** Advanced rendering tests (requires pygame display)

game.py Tests: 1/5 PASSED

- **Basic movement validation**
- **FAILED:** All integration tests due to import failures
- **FAILED:** State management tests
- **FAILED:** Game flow tests

DEFECT TRENDS & ANALYSIS

By Component

- **game.py:** 7 defects
- **shapes.py:** 1 defect
- **leaderboard.py:** 1 defect
- **All other components:** 0 defects

By Category

- **Dependencies/Integration:** 3 defects (D-31, D-35, D-38)
- **Game State Management:** 2 defects (D-32, D-33)
- **Input/Controls:** 2 defects (D-34, D-36)
- **Resources/Assets:** 2 defects (D-37, D-39)

Critical Path Failures:

1. Import failures block execution - Game cannot start without fixing
2. Missing game-over logic - Game never ends
3. State corruption - Failed games affect future sessions

Resolution Priority:

1. Fix import statements in game.py (D-31)
2. Implement game-over detection (D-32)
3. Add complete state reset between games (D-33)
4. Implement rotation cooldown (D-34)
5. Add leaderboard functionality (D-35)
6. Improve input handling (D-36)

Overall Assessment

Test Coverage: 18/24 (75%)

- **Passing Tests:** 18
- **Failing Tests:** 6
- **Critical Failures:** 4

Stability Rating: POOR

- **UI Components:** Good - Home and login work
- **Core Gameplay:** Broken - Critical issues prevent proper play
- **Integration:** Poor - Modules don't connect properly
- **User Experience:** Poor - Multiple frustrating bugs

Playability Status: NOT PLAYABLE

The game has critical issues that prevent normal gameplay:

1. May fail to start due to import errors
2. If starts, never properly ends (no game-over)
3. Controls are frustrating (rapid rotation)
4. State corruption breaks multiple sessions